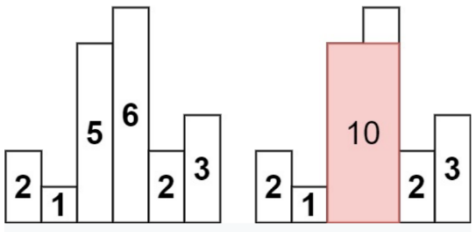
Lab 2 – Array, Sort, Stack

Instructions

- Choose one of the questions below.
- Read the problem and examples from the given link.
- Solve the problem
- Submit the solution or partial solution to Moodle.
- Make sure that your submission includes:
    - Question name
    - Link to the question website
    - Problem description (Copy&paste the question site is ok)
    - Explanation and thoughts about the solution including time and space complexity
    - The solution

1. **H-Index**
   https://leetcode.com/problems/h-index
   Given an array of integers citations where citations[i] is the number of citations a researcher received for their ith paper, return the researcher's h-index. Wikipedia definition: H-index is the maximum value of h such that the given researcher has published at least h papers that have each been cited at least h times.
   Examples: citations = [3,0,6,1,5] → H-index = 3,
   citations = [1,3,1] → H-index = 1.
2. **Sort array with squares**
   https://www.interviewbit.com/problems/sort-array-with-squares/
   Given a sorted array A containing N integers both positive and negative. You need to create another array containing the squares of all the elements in A and return it in non-decreasing order.
3. **Add One To Number**
   https://www.interviewbit.com/problems/add-one-to-number/
   Given a non-negative number represented as an array of digits, add 1 to the number (increment the number represented by the digits).
   MSB digit is first. Input number may have leading zeros. Output number may not have leading zeros. Input number may not be changed. Return the new number as output.
4. **Largest Rectangle in Histogram**
   https://leetcode.com/problems/largest-rectangle-in-histogram/
   Given an array of integers heights representing the histogram's bar height where the width of each bar is 1, return the area of the largest rectangle in the histogram.



5. **Max Rectangle in Binary Matrix**
   https://www.interviewbit.com/problems/max-rectangle-in-binary-matrix/
   Given a 2D binary matrix filled with 0's and 1's, find the largest rectangle containing **all ones** and return its area. Bonus if you can solve it in O(n^2) or less.

A =

1	1	1
0	1	1
1	0	0

Output = 4

6. **Remove Duplicate Letters**
   https://leetcode.com/problems/remove-duplicate-letters/
   Given a string s, remove duplicate letters so that every letter appears exactly once. Return the smallest in lexicographical order among all possible results.

Example1: Input: s = "bcabc" => Output: "abc"

Example2: Input: s = "cbacdcbc" => Output: "acdb"